Synthesis, nucleation kinetics, growth and characterization of Bis (Thiourea) cadmium nitrate nonlinear optical single crystals

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***Abstract*— Bis (Thiourea) cadmium nitrate (BTCN) was synthesized from aqueous solution and the nucleation parameters such as solubility, metastable zonewidth, induction period, interfacial energy were estimated. Single crystal and powder X-ray diffraction studies confirmed that PPHB crystallizes in monoclinic crystal system with space group Cc. The crystalline perfection of the grown crystal was analyzed by powder X-ray diffraction. Infrared and NMR spectroscopy techniques were used to elucidate the functional groups present in the compound. Mass spectral analysis provides additional confirmation of the formation of this new compound with a reasonable molecular ion peak at m/z 294.28. UV–Vis spectrum indicates that the crystal is transparent in the entire visible region with a lower cut off wavelength of 255 nm. The relative second harmonic generation efficiency was tested by Kurtz–Perry powder technique, which was about 1.21 times that of Potassium dihydrogen phosphate (KDP).**

Keywords:* *Crystal growth; Nucleation Kinetics; X-ray diffraction; Spectral analyses; Optical material; Thermal analysis

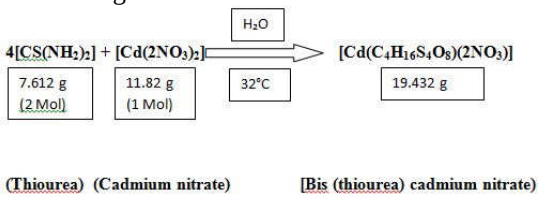
I. INTRODUCTION

Nonlinear optics (NLO) has emerged as one of the most attractive fields of current research in view of its vital applications in areas like optical modulation, optical switching, optical logic, frequency shifting and optical data storage for developing technologies in telecommunications and signal processing. Materials with large SONLO properties, short transparency cut off wavelengths and stable physicochemical performance are needed to realize many of the above applications [1]. Among the materials producing NLO effects, in particular the second harmonic generation (SHG), the first choice given is for those materials which possess high nonlinearity, shorter UV cutoff wavelength, large optical transparency window, ability to grow easily in large dimension, high laser damage threshold and good physicochemical stability. In recent years there has been considerable interest in the synthesis of semi-organic NLO materials with excellent second order optical nonlinearities [2-4]. Semi organic nonlinear optical crystals combine good thermal and mechanical properties of inorganic with high optical nonlinearity of purely organic compound [5-7]. Semiorganic crystals, which have the combined properties of both inorganic and organic crystals, are of interest presently. Among the semiorganic nonlinear optical materials, metal complexes of thiourea which have low UV cut off wavelengths, applicable for high power frequency conversion have received much attention. These materials can be used as better alternatives for KDP crystals in frequency doubling and laser fusion experiments due to their higher values of laser damage threshold. Growth of bulk single crystals of these materials has been a subject of perennial concern to enable it to be useful for device application. Bis thiourea cadmium nitrate (BTCN) is a potential NLO material amongst the metal complexes of thiourea. BTCN is 1.21 times more nonlinear than potassium dihydrogen phosphate (KDP). In the present investigation, bulk crystal of Bisthiourea cadmium nitrate (BTCN) was grown from water solvent and it's structural, spectral and optical properties have been studies in detail. In this present report, a systematic study have been made to estimate the solubility, metastable zonewidth, induction period (τ), interfacial energy (γ) for the bulk growth of BTCN crystal. The crystals grown by slow evaporation method have also been subjected to structural, spectral and optical studies.

II. EXPERIMENTAL

2.1 Synthesis of Bis (thiourea) cadmium nitrate

The starting reagents of commercially available thiourea ($CS(NH_2)_2$) and cadmiumnitrate ($Cd(NO_3)_2$) were taken in the stoichiometric ratio of 2:1. The calculated amount of cadmiumnitrate (11.82 gm) was dissolved in millipore water (100 ml) at 32°C, and then the appropriate amount of thiourea (7.612) was added to the solution. The solution was continuously stirred for about eight hours to achieve homogeneity of the mixed solution. The solution was filtered using filter paper and was allowed for slow evaporation to get the spontaneous nucleated crystals. The synthesized salt was further purified by successive recrystallization process in water and it was utilized for the crystal growth. The required quantity of thiourea and cadmiumnitrate was estimated from the following reaction:



2.2 Nucleation kinetics

2.2.1 Solubility of Bis (thiourea) cadmium nitrate

The solubility of Bis (thiourea) cadmium nitrate was determined at four different temperatures 30, 35, 40 and 45 ^{o}C. The solubility for 30 ^{o}C was determined by dissolving Bis (thiourea) cadmium nitrate salt in 100ml of an triplet distilled water taken in an air tight container maintained at the temperature with continuous stirring using ultracryostat with a magnetic stirrer at an accuracy of $\pm$ 0.01 ^{o}C. The saturation point was confirmed by the formation of undissloved salt at the bottom beaker. Then 5 ml of the saturated solution was pippeted out and poured in a Petridish. Then it was dried by keeping it in the hot air oven. After the evaporation of the solvent the Petridish was weighted. Then the different in the weight was taken into account. Then the different weight was repeated to estimate the solubility for different temperatures. The solubility graph is shown in Fig.1.

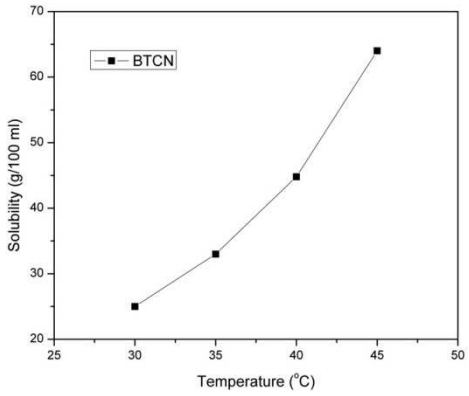


Fig. 1 Solubility of Bis (thiourea) cadmium nitrate

2.2.2 Metastable Zone Width

The saturated solution of Bis (thiourea) cadmium nitrate was prepared in accordance with the presently determined solubility data for the nucleation experiments. The studies were carried out in a constant temperature water bath controlled to an accuracy of $\pm$ 0.01 ^{o}C, provided with a cryostat for cooling below room temperature. A constant volume of 100 ml of solution was used in all the experiments. The solution was preheated to 5°C above the saturation temperature for one hour before cooling. It was continuously stirred to ensure homogeneous concentration and temperature through the entire volume of the solution. Metastable zone width of Bis (thiourea) cadmium nitrate was measured by the conventional polythermal method. In the polythermal method, the equilibrium-saturated solution is cooled from the overheated temperature until the first visible crystal is observed. Since the time taken for the formation of the first visible crystal after the attainment of critical nucleus is very small, the first crystal observed may be taken as the critical nucleus. The value of the metastable zone width depends not only on the temperature but also on the type of the crystal and its physio-chemical properties. The metastable zone width of Bis (thiourea) cadmium nitrate in aqueous solution as a function of temperature indicates that the metalstable zone width increase with temperature. Fig.2 shows the metastable zone width of Bis (thiourea) cadmium nitrate.

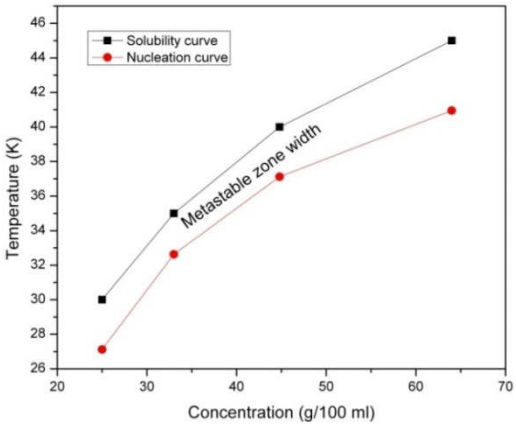


Fig. 2 Metastable Zone Width of Bis (thiourea) cadmium nitrate

2.2.3 INDUCTION PERIOD

To study the effect of supersaturation, induction period measurement was performed for various supersaturations at a temperature of 40 ^{o}C using isothermal method. The saturated solution was cooled to the desired temperature and by maintaining at that temperature the time taken for the formation of the first crystal was noted. Repeated trials were performed to ascertain the correctness of the observed results. Fig.3 shows the induction period v_s supersaturation(s)

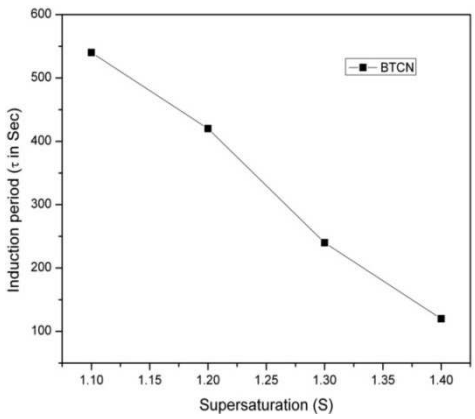


Fig. 3 Induction period vs. Supersaturation

2.2.4 INTERFACIAL TENSION

Interfacial tension of the crystal-solution interface is an important parameter involved in the theory of nucleation and growth kinetics. The classical homogeneous nucleation theory has been successfully tested for the nucleation of melts. Interfacial tension in the present investigation has been calculated on the basis of the classical theory of homogenous formation of spherical nuclei. Interfacial tension has been calculated at various constant temperature from graph drawn between In τ and 1/ (In S)^2 in Fig.4.

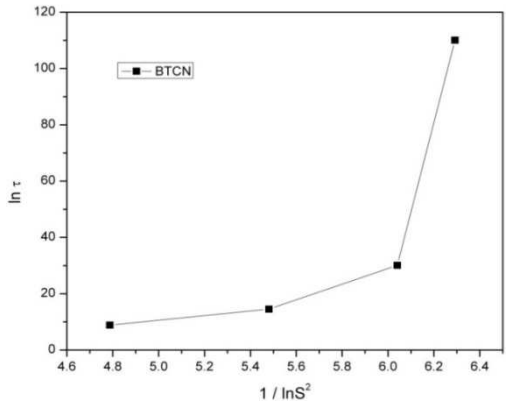


Fig. 4 Interfacial tension

$$\text{In } \tau = -\text{ In B} + \frac{16\pi\sigma^3 V^2 N_A}{3R^3T^3(\text{In S})^2} \qquad (1)$$

Where V is the molar volume of the crystal, N_A is the Avogadro's number, R is the gas constant, τ is induction period of BTCN solution at temperature T and B is constant. S is the supersaturation (S= C/ C*) where C is the actual concentration and C* is the equilibrium concentration.

$$S = 1 + \frac{\Delta H \Delta T}{RT^2} \qquad (2)$$

Where S can be calculated from the linear relation between ln χ_m and 1/T. ΔH is the enthalpy of dissolution ΔT is the supercooling and T is the absolute temperature.

Eq.(1) suggests a straight line for Inτ against 1/ (In S)^2 and therefore the slope is given by

$$m = \frac{16\pi\sigma^3 V^2 N}{3R^3T^3} \qquad (3)$$

Since In B weakly depends on temperature. Therefore, the interfacial tension is evaluated from

$$\sigma^3 = \frac{3\,R^3T^3\ N}{16\pi V^2 N} \qquad (4)$$

Table 1 Calculated growth parameters

Temperature ^{o}C	Solubility (gm/ 100ml)	Metastable Zone width (ΔT_C)	Supersaturation(S)	Induction period (τ) min
30	25	27.12	1.1	540
35	33	2.63	1.2	360
40	44.8	37.12	1.3	240
45	64	40.95	1.4	120

2.3 Crystal growth of Bis (thiourea) cadmium nitrate

The highly purified Bis (thiourea) cadmium nitrate salt was dissolved in water and solution was saturated at 35°C using a constant temperature bath with an accuracy of ± 0.01°C. Bis (thiourea) cadmium nitrate crystal was grown by employing solvent evaporation solution growth method. The optically good transparent Bis (thiourea) cadmium nitrate crystal with dimension of 15 x 9 x 4 mm^3 was harvested in a growth period of 4-5 weeks and is shown in Fig.5.

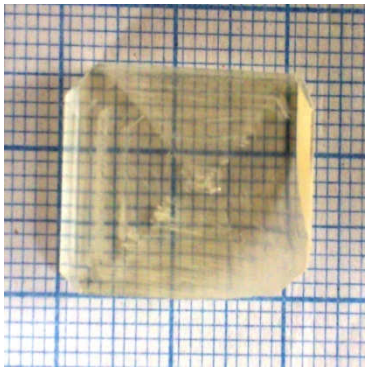


Fig. 5 Photograph of Bis (thiourea) cadmium nitrate

2.4 Characterization

Bruker Nonius CAD-4/MACH 3 single crystal X-ray diffractometer was used to estimate the lattice parameters of the grown crystal. Powder X-ray diffraction analysis was performed by using Bruker AXSCAD4 diffractometer with CuK_α (λ = 1.5405 Å) radiation to study the crystalline quality. FT-IR spectral technique was used to analyze the presence of functional groups in the title compound by using JASCO FTIR 410 spectrometer. Carbon and proton nuclear magnetic resonance spectroscopic methods were employed to elucidate the molecular structure of the compound using AM×400 NMR analyzer and Bruker AVANCE III 500 MHz spectrometers respectively. UV-Visible spectroscopy was employed to investigate the optical transmission property of the grown crystal using Perkin-Elmer Lambda 35 spectrophotometer. Mass spectral analysis was also carried out using a Finnigan MST 8230 mass spectrometer for confirmation of the formation of new, title compound. The energy dispersive spectroscopy analysis (EDS) was employed to find the stoichiometric composition, which was done using INCA 200 system connected to a LEO-Stereo scan 440 scanning electron microscope. Kurtz-Perry powder second harmonic generation test were performed on grown crystal using Q-switched Nd:YAG laser.

III. RESULTS AND DISCUSSION

3.1 Single crystal and powder X-ray diffraction studies

Figure Single crystal X-ray diffraction analysis was performed on Bisthiourea cadmium nitrate crystal to obtain the accurate cell parameters. The crystal structure was solved by direct method using SHELXS-97 program and the refinement was made by full-matrix least-square method. The grown crystal belongs to monoclinic crystal system with centrosymmetric space group $P2_1/C$. The estimated cell parameter values are a = 12.86 Å, b = 9.18 Å, c = 15.81 Å, $\alpha = \gamma = 90^o$, $\beta = 91.23^o$ and V = 1866 $Å^3$. It is also known that most of the Cd(II) in biological systems is not in the form of free Cd(II) ions, but is coordinated by the abundance of biological ligands [8]. Therefore the coordination chemistry of Cd(II) with ligands is of great interest. In this paper, we reported the synthesis and the crystal structure of Bis (thiourea) cadmium nitrate.

In the title compound, $[Cd(C_4H_{16}S_4O_8)](2NO)_3$, was prepared by reaction of cadmium nitrate and Bis (thiourea) in water solution. In the molecule of (I) (Fig.6), each Cd atoms is coordinated by four sulphurs atoms from four carbon atoms respectively. All the bond length and angle are in the normal range. Nitrate anion exists in the crystal lattice. The molecular structure and packing are stabilized by O-H...O and O-H...S inter hydrogen-bonding interactions (Fig.7) and crystallographic tables are listed.

The well defined Bragg's peaks at specific 2θ values for crystal planes and its corresponding miller indices were indexed in the recorded powder X-ray diffraction pattern (Fig.8). The sharp intensity peaks indicate the good crystallinity of the grown Bis (thiourea) cadmium nitrate single crystal.

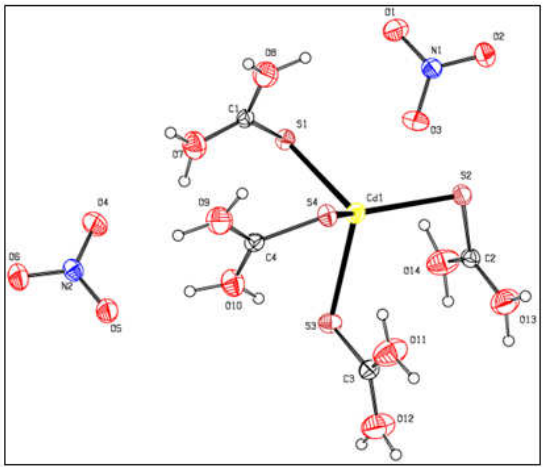


Fig. 6 Ortep diagram of Bis (thiourea) cadmium nitrate

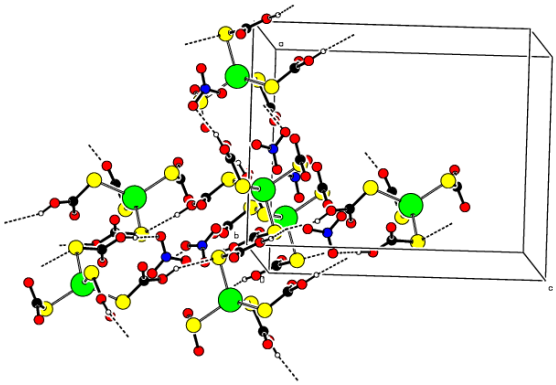


Fig. 7 Crystal Packing of Bis (thiourea) cadmium nitrate

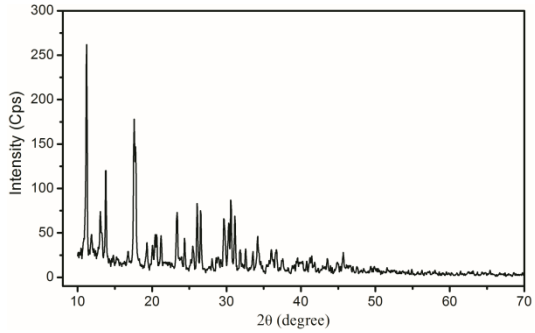


Fig. 8 Powder X-ray diffraction of Bis (thiourea) cadmium nitrate

Table 2 Crystal data and structure refinement for Bis (thiourea) cadmium nitrate

Identification code	shelxl
Empirical formula	$C_4 H_{16} Cd N_2 O_{14} S_4$
Formula weight	556.83
Temperature	293(2) K
Wavelength	0.71073 A
Crystal system, space group	monoclinic, P21/c
Unit cell dimensions	a = 12.8185(8) A alpha = 90 deg. b = 9.1543(5) A beta = 91.248(2) deg. c = 15.7702(11) A gamma = 90 deg.
Volume	1850.1(2) A^3
Z, Calculated density	4, 1.999 Mg/m^3
Absorption coefficient	1.699 mm^-1
F(000)	1112
Crystal size [mm]	0.24 x 0.20 x 0.18
Theta range for data collection	1.59 to 25.00 deg.
Limiting indices	-15<=h<=14, -10<=k<=7, -18<=l<=18
Reflections collected / unique	15931 / 3256 [R(int) = 0.0255]
Completeness to theta = 25.00	100.0 %
Refinement method	Full-matrix least-squares on F^2
Data / restraints / parameters	3256 / 16 / 291
Goodness-of-fit on F^2	1.076
Final R indices [I>2sigma(I)]	R1 = 0.0263, wR2 = 0.0747
R indices (all data)	R1 = 0.0295, wR2 = 0.0792
Extinction coefficient	0.0033(4)
Largest diff. peak and hole	0.457 and -0.496 e.A^-3

Table 3. Atomic coordinates (x 10^4) and equivalent isotropic displacement parameters (A^2 x 10^3) for Bis (thiourea) cadmium nitrate. U(eq) is defined as one third of the trace of the orthogonalized Uij tensor.

	x	y	z	U(eq)
O(8)	1078(2)	5997(3)	115(2)	55(1)
O(4)	-640(2)	8833(4)	-2284(2)	65(1)
O(3)	3648(2)	6853(3)	810(2)	54(1)
O(9)	1972(3)	7773(4)	-2219(2)	61(1)

O(12)	3726(3)	14158(4)	-560(3)	78(1)
C(3)	3504(3)	12759(4)	-462(2)	36(1)
O(13)	4512(3)	12361(4)	1814(2)	69(1)
Cd(1)	2535(1)	9595(1)	502(1)	35(1)
S(1)	777(1)	8549(1)	877(1)	32(1)
S(2)	3691(1)	9725(1)	1815(1)	38(1)
S(4)	3110(1)	8309(1)	-862(1)	35(1)
S(3)	2336(1)	12284(1)	-23(1)	39(1)
O(11	4195(3)	11801(4)	-713(3)	78(1)
N(2)	-585(2)	9552(3)	-2937(2)	36(1)
C(1)	669(2)	7312(3)	40(2)	28(1)
O(10)	1726(2)	10044(4)	-1677(2)	54(1)
O(7)	157(2)	7682(4)	-658(2)	57(1)
O(5)	240(2)	10221(3)	-3101(2)	50(1)
O(1)	2762(2)	5202(3)	1442(2)	49(1)
N(1)	3596(2)	5843(3)	1327(2)	36(1)
O(6)	-1350(2)	9640(3)	-3419(2)	60(1)
C(2)	3678(2)	11602(3)	1971(2)	31(1)
O(2)	4410(3)	5473(3)	1718(2)	67(1)
C(4)	2195(2)	8768(3)	-1638(2)	29(1)
O(14)	2850(3)	12262(3)	2238(2)	67(1)

Table 4. Bond lengths [A] and angles [deg] for Bis (thiourea) cadmium nitrate.

O(8)-C(1)	1.318(4)
O(8)-H(8B)	0.974(19)
O(8)-H(8A)	0.97(2)
O(4)-N(2)	1.226(4)
O(3)-N(1)	1.235(4)
O(9)-C(4)	1.319(4)
O(9)-H(9B)	0.98(2)
O(9)-H(9A)	0.98(2)
O(12)-C(3)	1.322(5)
O(12)-H(12A)	0.97(2)
O(12)-H(12B)	0.96(2)
C(3)-O(11)	1.313(5)
C(3)-S(3)	1.720(4)
O(13)-C(2)	1.304(4)
O(13)-H(13B)	0.97(2)
O(13)-H(13A)	0.96(2)
Cd(1)-S(2)	2.5232(9)
Cd(1)-S(1)	2.5301(8)
Cd(1)-S(4)	2.5738(9)
Cd(1)-S(3)	2.6079(9)
S(1)-C(1)	1.742(3)
S(2)-C(2)	1.737(3)
S(4)-C(4)	1.729(3)
O(11)-H(11B)	0.96(2)
O(11)-H(11A)	0.95(2)
N(2)-O(6)	1.230(4)
N(2)-O(5)	1.255(4)
C(1)-O(7)	1.313(4)
O(10)-C(4)	1.314(5)
O(10)-H(10B)	0.959(19)
O(10)-H(10A)	0.967(19)
O(7)-H(7A)	0.976(19)
O(7)-H(7B)	0.97(2)
O(1)-N(1)	1.237(4)
N(1)-O(2)	1.246(4)
C(2)-O(14)	1.299(5)
O(14)-H(14B)	0.97(2)
O(14)-H(14A)	0.95(2)
C(1)-O(8)-H(8B)	119(2)
C(1)-O(8)-H(8A)	125(3)
H(8B)-O(8)-H(8A)	114(4)
C(4)-O(9)-H(9B)	107(5)
C(4)-O(9)-H(9A)	124(4)
H(9B)-O(9)-H(9A)	127(6)
C(3)-O(12)-H(12A)	121(3)
C(3)-O(12)-H(12B)	124(4)
H(12A)-O(12)-H(12B)	114(5)

O(11)-C(3)-O(12)	117.6(4)
O(11)-C(3)-S(3)	123.5(3)
O(12)-C(3)-S(3)	118.9(3)
C(2)-O(13)-H(13B)	126(3)
C(2)-O(13)-H(13A)	117(3)
H(13B)-O(13)-H(13A)	117(4)
S(2)-Cd(1)-S(1)	109.65(3)
S(2)-Cd(1)-S(4)	122.25(3)
S(1)-Cd(1)-S(4)	107.17(3)
S(2)-Cd(1)-S(3)	105.53(3)
S(1)-Cd(1)-S(3)	110.46(3)
S(4)-Cd(1)-S(3)	101.21(3)
C(1)-S(1)-Cd(1)	97.22(10)
C(2)-S(2)-Cd(1)	98.96(11)
C(4)-S(4)-Cd(1)	106.13(11)
C(3)-S(3)-Cd(1)	106.62(12)
C(3)-O(11)-H(11B)	122(3)
C(3)-O(11)-H(11A)	125(4)
H(11B)-O(11)-H(11A)	112(5)
O(4)-N(2)-O(6)	119.7(3)
O(4)-N(2)-O(5)	119.9(3)
O(6)-N(2)-O(5)	120.3(3)
O(7)-C(1)-O(8)	120.0(3)
O(7)-C(1)-S(1)	119.8(3)
O(8)-C(1)-S(1)	120.1(2)
C(4)-O(10)-H(10B)	119(3)
C(4)-O(10)-H(10A)	119(2)
H(10B)-O(10)-H(10A)	121(4)
C(1)-O(7)-H(7A)	121(3)
C(1)-O(7)-H(7B)	117(4)
H(7A)-O(7)-H(7B)	121(5)
O(3)-N(1)-O(1)	120.7(3)
O(3)-N(1)-O(2)	118.2(3)
O(1)-N(1)-O(2)	121.0(3)
O(14)-C(2)-O(13)	119.5(3)
O(14)-C(2)-S(2)	121.1(3)
O(13)-C(2)-S(2)	119.3(3)
O(10)-C(4)-O(9)	119.4(3)
O(10)-C(4)-S(4)	123.5(2)
O(9)-C(4)-S(4)	117.2(3)
C(2)-O(14)-H(14B)	113(3)
C(2)-O(14)-H(14A)	111(4)
H(14B)-O(14)-H(14A)	137(5)

Table 5. Anisotropic displacement parameters (A^2 x 10^3) for Bis (thiourea) cadmium nitrate. The anisotropic displacement factor exponent takes the form: -2 pi^2 [h^2 a*^2 U11 + ... + 2 h k a* b* U12]

	U11	U22	U33	U23	U13	U12
O(8)	63(2)	52(2)	51(2)	-3(1)	-5(1)	-3(1)
O(4)	69(2)	80(2)	45(2)	28(2)	-11(1)	-16(2)
O(3)	63(2)	41(2)	58(2)	18(1)	-1(1)	-2(1)
O(9)	75(2)	63(2)	44(2)	-6(1)	-5(2)	-7(2)
O(12)	80(3)	55(2)	100(3)	7(2)	13(2)	-10(2)
C(3)	40(2)	31(2)	38(2)	-1(1)	-4(1)	-5(1)
O(13)	58(2)	51(2)	98(3)	4(2)	1(2)	-4(2)
Cd(1)	38(1)	38(1)	29(1)	-2(1)	-2(1)	-8(1)
S(1)	33(1)	37(1)	27(1)	-3(1)	2(1)	0(1)
S(2)	44(1)	27(1)	41(1)	-3(1)	-12(1)	2(1)
S(4)	36(1)	38(1)	32(1)	-3(1)	-3(1)	6(1)
S(3)	43(1)	33(1)	41(1)	1(1)	7(1)	1(1)
O(11)	63(2)	59(2)	113(3)	-2(2)	26(2)	-4(2)
N(2)	41(2)	40(2)	29(1)	-2(1)	-2(1)	4(1)
C(1)	23(1)	34(2)	26(2)	2(1)	2(1)	-4(1)
O(10)	55(2)	59(2)	47(2)	-3(1)	-7(1)	4(2)
O(7)	55(2)	70(2)	44(2)	0(2)	-5(1)	0(2)
O(5)	43(2)	62(2)	44(2)	10(1)	-4(1)	-7(1)
O(1)	50(2)	43(2)	56(2)	3(1)	5(1)	-6(1)
N(1)	46(2)	25(1)	36(2)	-5(1)	-7(1)	0(1)
O(6)	53(2)	76(2)	50(2)	13(1)	-18(1)	-9(1)
C(2)	35(2)	29(2)	29(2)	0(1)	-3(1)	0(1)
O(2)	63(2)	50(2)	86(2)	16(2)	-38(2)	-6(1)
C(4)	32(2)	32(2)	24(2)	0(1)	5(1)	-4(1)
O(14)	75(2)	47(2)	80(2)	-8(2)	19(2)	-5(2)

Table 6 Hydrogen coordinates (x 10^4) and isotropic displacement parameters (A^2 x 10^3) for Bis (thiourea) cadmium nitrate.

	x	y	z	U(eq)
H(10B)	1890(30)	10760(40)	-1250(20)	57(12)
H(10A)	1140(20)	10180(40)	-2070(20)	42(10)
H(8B)	1510(30)	5770(40)	610(19)	52(12)
H(8A)	130(40)	5300(40)	-350(20)	73(16)
H(7A)	60(30)	6990(40)	-1120(20)	63(13)
H(12A)	4340(30)	14460(60)	-860(30)	77(17)
H(11B)	4860(20)	12090(50)	-930(30)	69(14)
H(13B)	4560(40)	13410(20)	1800(30)	77(16)
H(14B)	2920(40)	13320(20)	2240(30)	65(13)
H(13A)	5110(30)	11820(50)	1630(30)	62(13)
H(11A)	4090(50)	10770(20)	-740(40)	89(18)
H(7B)	-40(50)	8700(30)	-720(50)	120(20)
H(14A)	2350(40)	11570(60)	2410(40)	130(30)
H(12B)	3250(40)	14940(50)	-440(40)	92(19)
H(9B)	2450(50)	6960(60)	-2120(60)	190(40)
H(9A)	1510(40)	7940(70)	-2720(30)	100(19)

Table 7. Torsion angles [deg] for Bis (thiourea) cadmium nitrate.

S(2)-Cd(1)-S(1)-C(1)	-133.72(10)
S(4)-Cd(1)-S(1)-C(1)	0.97(11)
S(3)-Cd(1)-S(1)-C(1)	110.39(11)
S(1)-Cd(1)-S(2)-C(2)	-107.35(12)
S(4)-Cd(1)-S(2)-C(2)	126.08(11)
S(3)-Cd(1)-S(2)-C(2)	11.62(12)
S(2)-Cd(1)-S(4)-C(4)	-169.47(11)
S(1)-Cd(1)-S(4)-C(4)	62.87(11)
S(3)-Cd(1)-S(4)-C(4)	-52.87(11)
O(11)-C(3)-S(3)-Cd(1)	19.3(4)
O(12)-C(3)-S(3)-Cd(1)	-161.1(3)
S(2)-Cd(1)-S(3)-C(3)	73.67(12)
S(1)-Cd(1)-S(3)-C(3)	-167.90(12)
S(4)-Cd(1)-S(3)-C(3)	-54.62(12)
Cd(1)-S(1)-C(1)-O(7)	-95.0(3)
Cd(1)-S(1)-C(1)-O(8)	86.5(3)
Cd(1)-S(2)-C(2)-O(14)	72.0(3)
Cd(1)-S(2)-C(2)-O(13)	-107.7(3)
Cd(1)-S(4)-C(4)-O(10)	34.5(3)
Cd(1)-S(4)-C(4)-O(9)	-145.9(2)

3.2 FTIR spectral analysis

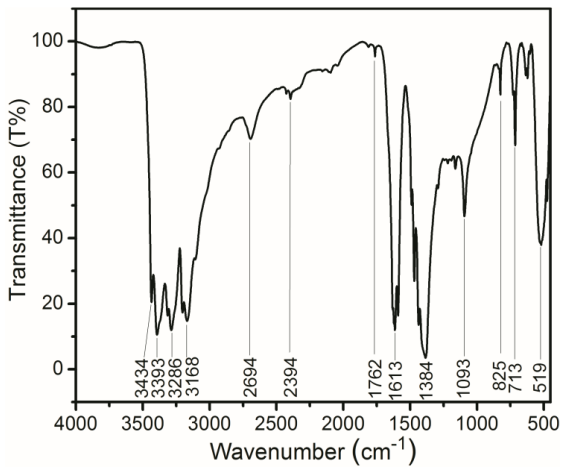


Fig. 9 FTIR spectrum of Bis (thiourea) cadmium nitrate

Fourier transform infrared spectrum of synthesized compound was recorded by KBr pellet method in the wavelength region 400-4000 cm^{-1}. This gives sufficient information on the structure of compound and is shown in Fig.9. In Bis (thiourea) cadmium nitrate, there are two possibilities by which the coordination of cadmium with thiourea can occur. The coordination with the cadmium may occur either through nitrogen or through sulfur of thiourea. The study of the spectra of metal complex Bisthiourea cadmium nitrate shows a shift in the frequency band in the low frequency region. The broad envelope positioned in between 2694 and 3434 cm^{-1} corresponds to the symmetric and asymmetric stretching modes of NH_2 grouping of cadmium coordinated thiourea. This indicates that nitrogen to cadmium bonds are not present in the coordination compounds. The absorption band observed at 1613 cm^{-1} in the spectrum of Bisthiourea cadmium nitrate and can be assigned to the NH_2 bending vibration. The absorption observed at 1490 and 1093 cm^{-1} in the spectrum of Bisthiourea cadmium nitrate and can be assigned to the N-C-N stretching vibration. The increase in the frequency can be attributed to the greater double bond character of the carbon to nitrogen bond on complex formation. The absorption band at 1384 cm^{-1} and can be assigned to the C=S stretching vibration. On coordination through sulfur, nature of vibration is slightly changed. The sharp and intense of Bisthiourea cadmium nitrate are absorption at 713 cm^{-1}. The lowering of frequency can be attributed to the reduced double bond character of the C=S bond on coordination. The absorption bands assigned to the particular vibrations indicate the presence of sulfur to cadmium bonds in Bisthiourea cadmium nitrate.

3.3 UV-Visible spectral studies

The optical transmittance spectrum of crystal recorded in the region 190-900 nm is shown in Fig.10. UV-visible spectrum reveals the electronic structure and optical band gap of the grown crystal. The higher transparency of the crystal, in the visible region is an important property for any nonlinear optical crystal for device fabrication. Bisthiourea cadmium nitrate showed good optical transmission of above 98% in the entire visible region with the cut-off wavelength of 255 nm and found suitable for optoelectronic applications. The sharp fall in the transmittance was observed for the grown Bis (thiourea) cadmium nitrate crystal and this corresponds to the fundamental absorption in the UV region. Absorption in the near ultraviolet region arises from π* and σ* electronic transitions associated within the sample. The study of the absorption edge is important especially in connection with the theory of electronic structure, which predicts that the band structure is mostly affected near the band extreme. Thus good transparency combined with low cutoff wavelength makes Bis (thiourea) cadmium nitrate's suitability for the nonlinear applications.

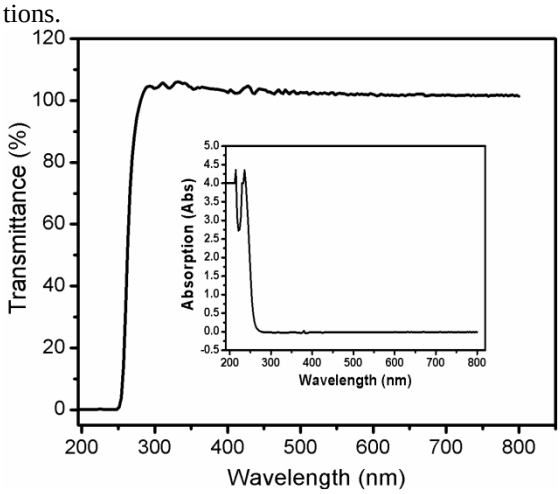


Fig. 10 UV-Visible spectrum of Bis (thiourea) cadmium nitrate

3.4 Proton and Carbon NMR spectral analysis

Proton nuclear magnetic resonance (NMR), a versatile technique is employed for the identification of organic compounds [9]. The spectrum is caused by protons in the different chemical environments in the given compound. Protons in identical chemical environment are called chemically equivalent protons and they give rise to a single signal in the spectrum. The inferences from NMR analysis provide very important deductions regarding the structure of a given compound as to, the functionality at a specific site, the groups at the neighboring carbon atoms and the number of protons causing that signal [10]. Carbon is the core element in organic chemistry and hence ^{13}C NMR plays an important role in determining the structure of unknown semiorganic molecules and the study of reaction and process. ^{1}H NMR and ^{13}C NMR spectrum was recorded by dissolving the sample in deuterated water and is shown in Fig.11 and Fig.12. The singlet peak at δ = 4.686 ppm are due to proton in NH_2 group of thiourea ring. The intense signal at δ = 7.427 ppm is assigned to proton in NH_2 group of thiourea. The carbon produces and its characteristic signal at 179.26 ppm. The characteristic functional groups of Bisthiourea cadmium nitrate compound were confirmed through the proton and carbon NMR intensities.

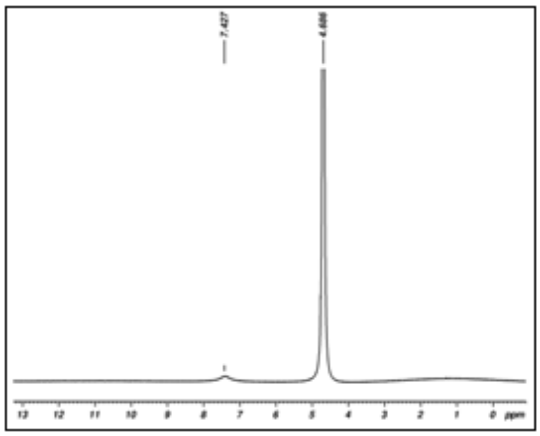


Fig. 11 Proton NMR spectrum of Bis (thiourea) cadmium nitrate

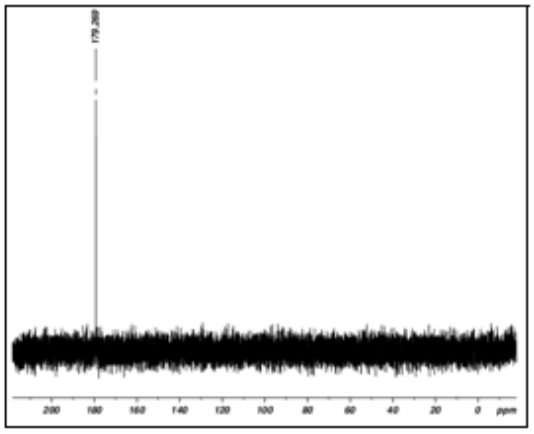


Fig. 12 Corbon NMR spectrum of Bis (thiourea) cadmium nitrate

3.5 Mass spectral analysis

Mass spectrometry was carried out for Bisthiourea cadmium nitrate in the mass range 50–350 amu is shown in Fig.13. The molecular weight of Bisthiourea cadmium nitrate is observed from the molecular ion peak at m/z 294.28. The base peak at m/z 59.67 and the fragmentation peak observed at m/z 78.06. From the results of ^{1}H-NMR, ^{13}C-NMR and Mass spectra the structure of a prepared compound was confirmed.

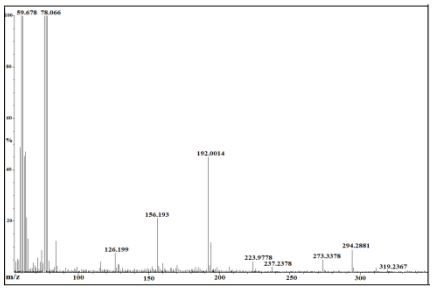


Fig. 13 Mass spectral analysis of Bis (thiourea) cadmium nitrate

3.6 Energy Dispersive Spectroscopy analysis

The stoichiometric composition of the as grown Bisthiourea cadmium nitrate single crystal was determined by energy dispersive spectroscopy analysis (EDS) using XRD INCA 200 system connected to a LEO-stereo scan 440 scanning electron microscope. The EDS pattern of Bisthiourea cadmium nitrate is given in Fig.14, which reveals the presence of Cd in the grown crystal. From the characteristic peaks of these elements, the presence of sulfur, carbon, oxygen and nitrogen were also detected. From the chemical and EDS analysis, the empirical formula for the crystal was determined to be Bis (thiourea cadmium nitrate).

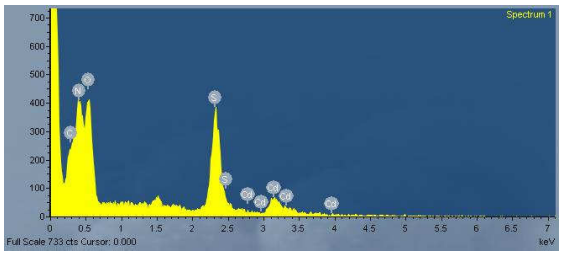


Fig. 14 EDS spectral analysis of Bis (thiourea) cadmium nitrate

3.7 Second harmonic generation test

The NLO property of the grown BTCN crystal was studied by Kurtz–Perry powder technique with KDP crystal as reference material [11]. Finely powdered crystal of BTCN was densely packed in a micro capillary tube of uniform bore. The second harmonic generation (SHG) measurement was carried out using 1064 nm Q-switched mode locked Nd:YAG laser. The pulse width of 8 ns and10 Hz repetition rate with pulse energy of 2.8 mJ/pulse laser was made to fall normally on the sample cell. An intense green light (λ = 532 nm) emission from the sample was detected by a photomultiplier tube which confirmed the second harmonic property of BTCN crystal. KDP powdered sample with identical size was used as reference material in the SHG measurement. The SHG signal voltages for BTCN and KDP were measured to be 29 mV and 24 mV respectively. Thus, SHG efficiency of Bisthiourea cadmium nitrate was found to be 1.21 times higher than that of KDP crystal.

IV. CONCLUSION

The fundamental nucleation parameters of the Bis (thiourea) cadmium nitrate single crystal were determined on the basis of the classical nucleation theory. Investigation on the fundamental growth parameters for the aqueous solution growth of Bisthiourea cadmium nitrate shows that the experimentally determined interfacial tension using the induction period values are good is agreement with the reported theoretical values estimated from solubility and induction period data. The induction period was calculated at different supersaturation ratio using the experimentally determined induction period values. Single crystal of Bisthiourea cadmium nitrate was grown from Millipore water solvent and its lattice parameters were obtained from single crystal X-ray diffraction analysis. The crystallinity perfection of the grown crystal was studied by powder X-ray diffraction studies. The presence of functional groups was confirmed by FTIR and carbon & proton NMR spectral analyses. UV cut-off wavelength (255 nm), transparency nearly 98% over the entire range of UV–visible region showed that Bisthiourea cadmium nitrate is a good candidate for nonlinear optical applications. Mass spectral analysis provides additional confirmation of the formation of this new compound with a reasonable molecular ion peak at m/z 294.28. The relative second harmonic generation efficiency of the grown crystal was found to be 1.21 times that of KDP crystal.

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